

How to run multiple containers from a single image, with a single command?

How to run multiple containers from a multiple image, with a single command?

solution: docker compose

How to run multiple containers from a single/multiple images, in to multiple VMs with a single command?

solution: container orchestration

- docker swarm
- kubernetes

How to run multiple containers from a given image?

Why we need more than one container for a single application? HA

We look for an automation tool

docker-compose tool (Single Node) helps to deploy multiple containers from single/multiple images with a single command.

We also need to write a script and call that as docker-compose file.

We write the compose file with extension yaml

service: group of containers from a single image only.

docker stack: group of services

docker-compose tool:

- helps to deploy multiple containers but on a single vm only.
- Dynamic nature of the container with port keeps changing is not possible.
- Is not right tool for production deployment, this can be used only for dev use cases.

service: myapp



How do execute /run docker-compose file in a single node

Using docker-compose utility we can run docker-compose file

We must install docker-compose utility on your VM.

step1: Install docker-compose utility (help us to run docker-compose file)

```
sudo curl -L
```

```
https://github.com/docker/compose/releases/download/1.23.2/docker-
```

```
compose-`uname -s`-`uname -m`-o /usr/local/bin/docker-compose
```

```
sudo chmod +x /usr/local/bin/docker-compose
```

docker-compose -v

step2: docker-compose means we are talking about multiple container (service)

service: group of containers from a single image.

step3: we can write multiple services in a single-compose file.

```
docker service create --name myapp --replicas=5 -p 9080:80 nginx:latest
```

```
docker service create --name pyapp --replicas=5 -p 9081:3000
```

```
lerndevops/samplepyapp:v1
```

```
docker service ps <service name> # spread across cluster
```

```
docker ps # container running on specific node
```

```
docker service rm <service name>
```

How to write docker-compose file?

```
nano docker-compose.yml
```

services:

```
myapp:    # this is the service name can be anything
```

```
image: nginx:latest
```

```
ports:    # publish the port
```

```
- 80      # -P option
```

```
# - 9080:80    # -p option
```

volumes:

```
# - src(external storage path):target(container path)
```

```
- /root/nginxlogs:/var/log/nginx
```

```
- /root/nginxconf:/usr/share/nginx/html
```

```
docker-compose -f docker-compose.yml up -d # this would create 1 cont.
```

```
docker-compose -f docker-compose.yml up --scale <servicename>=5 -d
```

```
docker-compose -f docker-compose.yml down
```

In docker-compose file we write service: group of containers from single image.
We can write multiple services in a single docker-compose file.

```
nano docker-compose.yml
```

```

version: '3.7'
volumes:
  nginxconf:
  tomcatlogs:
networks:
  mynet:
services:
  web: ## service name can any name of your choice
    image: nginx
    ports: ## the ports which are exposed inside the cont
      - "80"
      - "443"
    volumes:
      - "nginxconf:/usr/share/nginx/html"
    networks:
      - mynet
  app: ## service name can any name of your choice
    image: tomcat
    ports: ## the ports which are exposed inside the cont
      - "8080"
      - "8443"
    volumes:
      - "tomcatlogs:/usr/local/tomcat/logs"
    networks:
      - mynet
~
~
~
~

```

version: using docker-compose client can talk to the docker-engine

services: It not a single container, group of containers from a single image are called service (web).

web: name of the service

image: image used by service

ports: image run on which port (Array)

volume: volume-mount

docker-compose up -d # up bring up the containers

```

root@node1:~# docker-compose up -d
Creating network "root_mynet" with the default driver
Creating volume "root_nginxconf" with default driver
Creating volume "root_tomcatlogs" with default driver

```

docker-compose up --scale web=4 --scale app=6 -d

docker container ls # docker ps

It's a vertical scaling happening on the single VM.

docker-compose down

Customized file name?

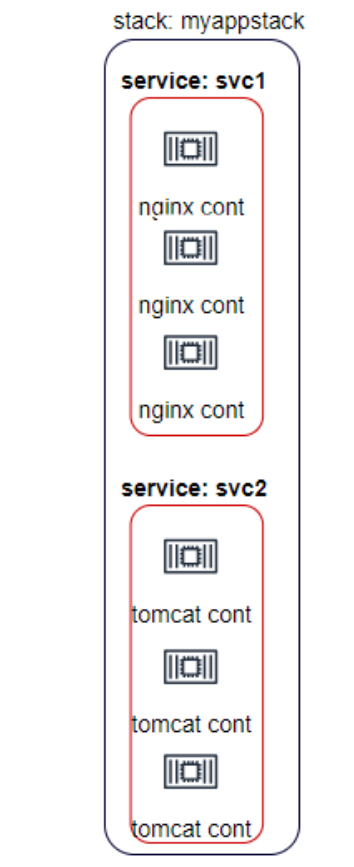
How docker-compose file identified docker-compose. yml?

docker-compose utility by default tries to read docker-compose. yaml, if we give some other name (mycompose. yaml) in that case, we have to provide the full path.

```
docker-compose up --file mycompose. yaml --scale web=2 --scale app=3 -d
```

```
docker-compose up --file mycompose. yaml down
```

Stack:



version: "3.7"

services:

svc1: # service name / can be any name of your choice

image: nginx:latest

ports:

- "80" # -P

```
# - "9080:80" # -p
```

```
deploy:
```

```
  replicas: 4
```

```
svc2: # service name / can be any name of your choice
```

```
  image: tomcat:8.5.40
```

```
  ports:
```

```
    - "8080" # -P
```

```
deploy:
```

```
  replicas: 4
```

```
docker stack deploy -c compose.yaml myappstack
```

Limitation:

- It brings up the container in the single node only (SPF), can't achieve HA.
- Not suitable for production environment.

compose utility is creating multiple containers in the same node, what happens if my node goes down? We are creating multiple containers for HA, but what will happen if my node goes down? Will I get the HA? No, its SPF. It's helpful in development but not recommended for production environment.

It just provides feature of replicas and scaling up and scaling down.

Solution: Container Orchestration tool

difference between Dockerfile and docker-compose?

dockerfile creates images.

docker-compose file creates containers from a given image.